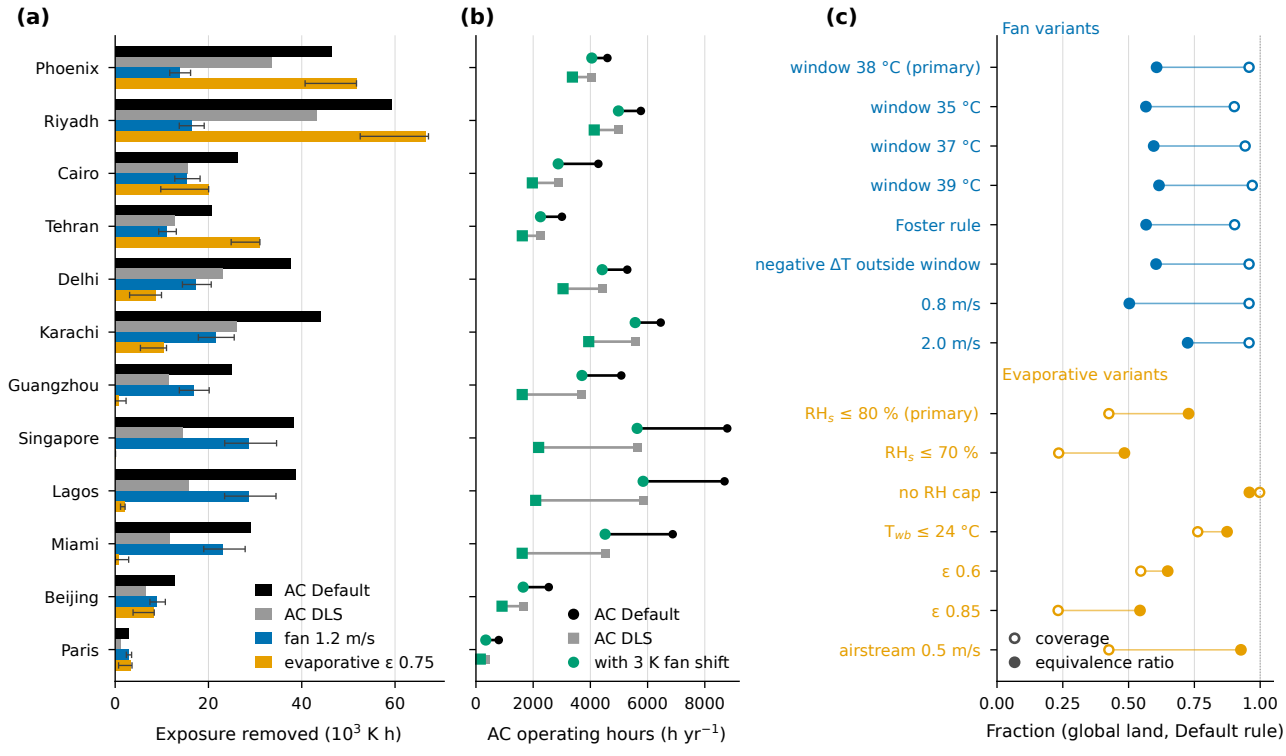
(a)
Phoenix
Riyadh
Cairo
Tehran
Delhi
Karachi
Guangzhou
Singapore
Lagos
Miami
Beijing
Paris
AC Default
AC DLS
fan 1.2 m/s
evaporative ε 0.75
0
20
40
60
Exposure removed (10^3 K h)
(b)
AC Default
AC DLS
with 3 K fan shift
0
2000
4000
6000
8000
AC operating hours (h yr^{-1})
(c)
Fan variants
window 38 °C (primary)
window 35 °C
window 37 °C
window 39 °C
Foster rule
negative ΔT outside window
0.8 m/s
2.0 m/s
Evaporative variants
RH_s ≤ 80 % (primary)
RH_s ≤ 70 %
no RH cap
T_{wb} ≤ 24 °C
ε 0.6
ε 0.85
airstream 0.5 m/s
coverage
equivalence ratio
0.00
0.25
0.50
0.75
1.00
Fraction (global land, Default rule)